

In Panel C of Figure 2.8, I show indifference curves for a more risk-averse investor with a risk aversion level of $\gamma = 7$. (Note that these indifference curves are steeper than the ones in Panel B.) The tangency point is marked in the triangle and is closer to the 100% bonds point on the mean-variance frontier. For the $\gamma = 7$ asset owner, a 32% equities position is optimal.

So what risk aversion level corresponds to a 60% equities position? Panel D of Figure 2.8 plots the equities weight as a function of risk aversion. A 60%/40% equities–bond position corresponds to a risk aversion level of $\gamma = 2.8$. Thus, typical pension funds—like most individual investors—are moderately risk averse. This method of backing out risk aversion from portfolios that are observed is called *revealed preference*.

3.4. THE MARKET'S RISK AVERSION

The *capital allocation line* (CAL) describes portfolios that can be chosen when there is one risky asset, say equities, and a risk-free asset. For now, interpret the latter as U.S. T-bills, even though these have a tiny bit of (sovereign) risk (see chapter 14), which we will ignore for the time being. We assume that a period is known—say, a quarter—so that we know what the risk-free rate is. We denote the risk-free rate by r_f and the expected return and standard deviation of the risky asset as $E(r)$ and σ , respectively.

The CAL is described in portfolio mean-standard deviation space ($E(r_p), \sigma_p$) by the following line:

$$E(r_p) = r_f + \frac{E(r) - r_f}{\sigma} \sigma_p. \quad (2.7)$$

The CAL traces out all the possible portfolios of a risk-free asset and a risky asset. Figure 2.9 plots the CAL for a risk-free rate of $r_f = 1\%$ and U.S. equities using Ibbotson data over January 1926 to December 2011. U.S. equities are marked in the square.

The slope of the CAL is the Sharpe ratio, named after William Sharpe, one of the founders of the capital asset pricing model and winner of the 1990 Nobel Prize:

$$\text{Sharpe Ratio} = \frac{E(r) - r_f}{\sigma}. \quad (2.8)$$

In Figure 2.9, the Sharpe ratio for U.S. equities was 0.53, using the average equity return over the sample and a 1% risk-free rate.¹⁸ The Sharpe ratio is actually a zero-cost trading strategy: in the context of Figure 2.9, it is long equities and short T-bills, divided by the risk of equities. It is the reward (mean excess return)

¹⁸ Note that “risk-free” rates move around in data. The empirical Sharpe ratio using T-bills as the risk-free asset is $0.0766/0.1918 = 0.40$.